Daniel Tamor Liu Citron

Curriculum Vitae

daniel.t.citron@gmail.com

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PROFESSIONAL EXPERIENCE

NYU Langone Health — New York, NY

Senior Research Scientist — 2022-Present

Facebook, Inc — Contracted through PRO Unlimited — Seattle, WA

Visiting Research Scientist — 2020 – 2021

University of Washington — Seattle, WA

Postdoctoral Researcher — 2017 – 2020

EDUCATION AND TRAINING

Cornell University — Ithaca, NY

Ph.D. Theoretical Physics; Experimental Physics Minor — 2011 – 2017

M.S. Physics — 2014

University of Chicago — Chicago, IL

B.A. Physics with Honors — 2009

TECHNICAL SKILLS

Languages: Python, SQL, R, Mathematica

Tools & Platforms: Git, VSCode, Claude Code

PUBLICATIONS

Citron, D.T., Kim, H.Y., Kasujja, R., Musanje, K., Mwalili, S.M., Gathungu, D.K., Wambui, P., Chambwe, C., Platais, I., Assefa, F.B. and Jeetoo, M., 2026. Cost-Effectiveness and Impact of Depression Treatment Implementation on HIV Outcomes in Western Kenya: A Mathematical Modelling Study. Journal of the International AIDS Society, 29(9), p.e70186.

Citron, D.T., Kim, H.Y., Kasujja, R., Mwalili, S., Gathungu, D., Chambwe, C., Platais, I., Assefa, F.B., Braithwaite, R.S. and Bershteyn, A., 2026. How has mental health impacted the HIV epidemic and how has the HIV epidemic impacted mental health in Western Kenya? A mathematical modelling study. BMJ Global Health, 11(8), p.e020217.

Bansi-Matharu, L., **Citron, D.T.**, Martin-Hughes, R., Moolla, H., Stover, J., Pickles, M., Mangal, T., Smith, J., Mugurungi, O., Kubyana, M.S. and Taramusi, I., 2026. HIV diagnosis and treatment status in individuals dying from HIV in Zimbabwe, Malawi, and South Africa: A model comparison analysis. PLoS medicine, 23(9), p.e1004864.

Milali, M.P., **Citron, D.T.**, Bhamidipati, K., Yamamoto, N., Osei-Ntansah, A., Platais, I., Ferrara, G., Ngcamphalala, C., Dlamini, S.G., Ginindza, N. and Bershteyn, A., 2026. Projected burden of hypertension-associated cardiovascular disease in people living with HIV versus HIV-negative adults in Eswatini. medRxiv, pp.2026-07.

You, S., Kim, H.Y., Phillips, A.N., **Citron, D.T.**, Kaftan, D., Platais, I., Bansi-Matharu, L., Cambiano, V., Nichols, B.E., Jo, Y. and Braithwaite, R.S., 2026, June. Cost-Effectiveness of Differentiated Service Delivery for HIV Treatment: A Combined Mathematical Modeling Study of Four African Settings. In Open Forum Infectious Diseases (Vol. 13, No. 6, p. ofag257). US: Oxford University Press.

Mahmud, Ayesha S., Meng-Chun Chang, **Daniel T. Citron**, Kenth Engø-Monsen, Abdullah Abu Sayeed, Sazid Ibna Zaman, Didar Uddin et al. "Identifying malaria elimination strategies in the presence of human movement in Bangladesh." *Communications Medicine* 5, no. 1 (2025): 461.

Stover, John, Emily Sonneveldt, Yvonne Tam, Katherine C. Horton, Andrew N. Phillips, Jennifer Smith, Rowan Martin-Hughes et al. "Effects of reductions in US foreign assistance on HIV, tuberculosis, family planning, and maternal and child health: a modelling study." *The Lancet Global Health* 13, no. 10 (2025): e1669-e1680.

Mulenga, Lloyd B., Kebby Musokotwane, Suilanji Sivile, Khozya D. Zyambo, Roma Chilengi, Kennedy Lishimpi, George Sinyangwe et al. "Impacts of US Bilateral Aid Disruptions on HIV Resurgence in Zambia: A Mathematical Modeling Study." *Open Forum Infectious Diseases* vol. 12, no. 9, p. ofaf511. US: Oxford University Press, 2025.

Bansi-Matharu, Loveleen, Haroon Moolla, **Daniel T. Citron**, John Stover, Michael Pickles, Rowan Martin-Hughes, Marie-Claude Boily et al. "Identifying gaps in the HIV treatment cascade in Africa: a model comparison study." *The Lancet Global Health* 13, no. 6 (2025): e1006-e1019.

Chowdhury, MD Tuhin, Anna Bershteyn, Masabho Milali, **Daniel T. Citron**, Sulani Nyimbili, Godfrey Musuka, and Diego F. Cuadros. "Assessing regional variations and sociodemographic barriers in the progress toward UNAIDS 95-95-95 targets in Zimbabwe." *Communications Medicine* 5, no. 1 (2025): 106.

Gathungu, Duncan K., Viona N. Ojiambo, Mark E. Kimathi, David Kaftan, Hae-Young Kim, **Daniel T. Citron**, Ingrida Platais et al. "Mitigating HIV risk associated with widow cleansing and wife inheritance using combined biomedical and structural interventions in western Kenya: a mathematical modeling study." *BMC Medicine* 23, no. 1 (2025): 88.

Taramusi, Isaac, John Stover, Robert Glaubius, Tsitsi Apollo, Getrude Ncube, Owen Mugurungi, Ngwarai Sithole et al. "HIV incidence and prevalence projections for Zimbabwe: Findings from five mathematical models." *African Journal of AIDS Research* 24, no. 1-2 (2025): 44-52.

Adam Akullian, Ruth Akulu, Gambo Aliyu, Florence Anam, Anne-Claire Guichard, Helen Ayles, Rachel Baggaley, et al. "The HIV response beyond 2030: preparing for decades of sustained HIV epidemic control in eastern and southern Africa." *The Lancet* (2024).

Diego F. Cuadros, Tuhin Chowdhury, Masabho Milali, **Daniel T. Citron**, Sulani Nyimbili, Natalie Vlahakis, Theodora Savory et al. "Geospatial patterns of progress towards UNAIDS '95-95-95'targets and community vulnerability in Zambia: insights from population-based HIV impact assessments." *BMJ Global Health 8*, no. 10 (2023): e012629.

Sean L. Wu, John M. Henry, **Daniel T. Citron**, Doreen Mbabazi Ssebuliba, Juliet Nakakawa Nsumba, Héctor M. Sánchez C, Oliver J. Brady et al. "Spatial dynamics of malaria transmission." *PLOS Computational Biology* 19, no. 6 (2023): e1010684.

Shankar Iyer, Brian Karrer, **Daniel T. Citron**, Farshad Kooti, Paige Maas, Zeyu Wang, Eugenia Giraudy, Ahmed Medhat, P. Alex Dow, and Alex Pompe. "Large-scale measurement of aggregate human colocation patterns for epidemiological modeling." *Epidemics* 42 (2023): 100663.

Daniel T. Citron, Carlos A. Guerra, Guillermo A. García, Sean L. Wu, Katherine E. Battle, Harry S. Gibson, and David L. Smith. "Quantifying malaria acquired during travel and its role in malaria elimination on Bioko Island." *Malaria Journal* 20, 359 (2021).

Sean L. Wu, Héctor M. Sánchez C., John M. Henry, **Daniel T. Citron**, Qian Zhang, Biyonka Liang, Kelly Compton, Amit Verma, Derek A. Cummings, Thomas W. Scott, Anne Wilson, Steven Lindsay, Catherine Moyes, Penny Hancock, Tanya Russell, Thomas R. Burkot, John Marshall, Samson Kiware, Robert C. Reiner Jr., & David L. Smith. "Vector bionomics and vectorial capacity as emergent properties of mosquito behaviors and ecology." *PLoS computational biology* 16, no. 4 (2020): e1007446.

Daniel T. Citron, Carlos A. Guerra, Andrew J. Dolgert, Sean L. Wu, John M. Henry, and David L. Smith. "Comparing metapopulation dynamics of infectious diseases under different models of human movement." *Proceedings of the National Academy of Sciences* 118, no. 18 (2021).

Carlos A. Guerra, **Daniel T. Citron**, Guillermo A. García, and David L. Smith. "Characterising malaria connectivity using malaria indicator survey data." *Malaria journal* 18, no. 1 (2019): 1-12.

Carlos A. Guerra, Su Yun Kang, **Daniel T. Citron**, Dianna EB Hergott, Megan Perry, Jordan Smith, Wonder P. Phiri, José O. Osá Nfumu, Jeremiás N. Mba Eyono, Katherine E. Battle, Harry S. Gibson, Guillermo A. García, & David L. Smith. "Human mobility patterns and malaria importation on Bioko Island." *Nature communications* 10, no. 1 (2019): 2332.

Daniel T. Citron and Samuel F. Way. "Network assembly of scientific communities of varying size and specificity." *Journal of Informetrics* 12, no. 1 (2018): 181-190.

Daniel T. Citron and Paul Ginsparg. "Patterns of Text Reuse in a Scientific Corpus." *Proceedings of the National Academy of Sciences* 112, no. 1 (2015): 25-30. DOI:10.1073/pnas.1415135111

Mark L. Rivers, **Daniel T. Citron**, and Yanbin Wang. "Recent developments in computed tomography at GSECARS." In *Developments in X-Ray Tomography VII*, vol. 7804, p. 780409. International Society for Optics and Photonics, 2010.

Xiang Cheng, German Varas, **Daniel T. Citron**, Heinrich M. Jaeger, and Sidney R. Nagel. "Collective behavior in a granular jet: Emergence of a liquid with zero surface tension." *Physical review letters* 99, no. 18 (2007): 188001.

CONFERENCES AND WORKSHOPS

Epidemiological impacts of abruptly de-implementing adolescent and young adult HIV services due to bilateral aid cuts in Zambia: a mathematical modeling study
Science of Dissemination and Implementation in Health — December 15-16, 2025
Poster Presentation

Estimating the negative impact of PEPFAR's cancellation on HIV health outcomes: a mathematical modeling study
MIDAS Network Meeting — November 3-5, 2025
Oral Presentation

Changes in social contact time following digital SARS-CoV-2 post-exposure notification: A retrospective case-control study at a US university.
NetMob 2025 — October 8-10, 2025
Oral Presentation

Modeled estimates of disease burden attributable to interactions between HIV and depression in Kenya
CROI March 3-6, 2024
Poster Presentation

Impact of universal screening and treatment for depression on HIV transmission and mortality in a high-prevalence region of Kenya: a mathematical modeling study
Epidemics 9 November 28-December 1, 2023
Poster Presentation

Changes in social interactions following digital exposure notifications: a case-control study among mobile phone contact tracing application users
Epidemics 9 November 28-December 1, 2023
Poster Presentation

Impact of universal screening and treatment for depression on HIV transmission and mortality in a high-prevalence region of Kenya: a mathematical modeling study
Global Mental Health Without Borders October 30-31, 2023
Contributed Presentation

Quantifying how human mobility drives spatial transmission dynamics of malaria with mathematical models
Strathmore International Mathematics Conference July 10-14, 2023
Contributed Presentation

Using mobile phone contact tracing app data to measure how behavior changes in response to notifications of exposure risk
MIDAS Annual Meeting September 7-9, 2022
Contributed Presentation

Activity Space Maps: a novel human mobility data set for quantifying time spent at risk
Epidemics November 30-December 3, 2021
Poster Presentation

Activity Space Maps: a novel human mobility data set for quantifying time spent at risk
KDD Workshop on Data-driven Humanitarian Mapping August 15, 2021
Contributed Presentation

Metapopulation Models of Infectious Disease Dynamics are Sensitive to Underlying Host Movement Models
Epidemics December 3-6, 2019
Contributed Presentation

Quantifying malaria acquired during travel and its role in malaria elimination on Bioko Island
ASTMH November 20-24, 2019
Poster Presentation

Sensitivity of Metapopulation Models of Infectious Disease Dynamics to Underlying Host Mobility Networks

NetSci 2019 May 27-31, 2019
Poster Presentation

Applied Simulation Modeling for Interrupting Malaria Transmission on Bioko Island

7th Annual Disease Modeling Symposium April 15-17, 2019
Invited Presentation

Agent-Based Modeling for Malaria Policy

African Health Economics and Policy Association Conference March 11-14, 2019
Invited Presentation

Network Analysis of Mosquito Habitats for Controlling Vector-Borne Pathogens

NetSci 2018 June 11-15, 2018
Poster Presentation

Contact Network Heterogeneity and Persistence of Endemic Disease

NetSci 2017 June 19-23, 2017
Contributed Presentation

Network Assembly in Scientific Collaboration Networks Northwestern U.

International Conference on Computational Social Science June 22-26, 2016
Poster Presentation

Network Analysis of ArXiv Santa Fe Institute

SFI Complex Systems Summer School 2015 July 3, 2015
Contributed Presentation

Moment Closure Analysis of SIRS Disease Model on Heterogeneous Networks

APS March Meeting 2015 March 2-6, 2015
Contributed Presentation

Accounting for Fluctuations in Stochastic SIRS Model on Networks U. of Pittsburgh

International Workshop on Advances in Discrete Networks December 12-14, 2014
Poster Presentation

AWARDS & FELLOWSHIPS

NSF Graduate Research Fellowship (Cornell University) 2012

TEACHING

Workshop on Mathematical Models of Infectious Disease Dynamics

Strathmore University July 2023
University of South Africa January 2024
Emirates Aviation University February 2024

Instructor Cornell University

Physics GRE Preparation Short Course Spring 2013, 2014, 2015, 2016

Teaching Assistant — Cornell University

Electricity and Magnetism (honors sequence)	Spring 2016
Mechanics and Special Relativity (honors sequence)	Fall 2014, 2015
Introduction to Electricity and Magnetism	Spring 2012
Mechanics and Heat	Fall 2011

OUTREACH AND SERVICE

Scientist-in-Residence, New York Academy of Sciences — New York, NY

5th Grade Science Classroom	Fall 2022 – Spring 2023
10th Grade Biology Classroom	Fall 2023 – Spring 2024

Pacific Science Center — Seattle, WA

Science Communication Fellow	Spring 2018 – 2020

Destination Imagination — Central New York

Board Member and Volunteer	Spring 2015 – Summer 2017

Graduate & Professional Students Assembly — Cornell University

Chair, Faculty Awards Committee	Fall 2014 – Fall 2015
Physics Field Representative	Fall 2013 – Spring 2016

Cornell Center for Materials Research Outreach — Cornell University

Volunteer	Summer 2014 – Summer 2017

University of Chicago Scavenger Hunt — University of Chicago

Judge (event organizer)	2009 – 2015